Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3311-WE01

Title:

MEng/BEng: Geotechnics 3

Time Allowed:	2 hours	
Additional Material provided:	Yes - Formula Sheet included as part of exam paper	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Q1

(a) A deposit of sandy silt is at a specific volume of 1.72 and the specific gravity of the sandy silt grains is 2.73. Show that the bulk unit weight of this soil is 15.6kN/m^3 when the sandy silt is in a dry condition and 19.7 kN/m^3 when it is in a saturated condition (assume that the density of water is 1000 kg/m^3). Justify your answer. [15%]

(b) In a deposit of silty clay, the diameter of the largest "throats" connecting between voids is approximately 0.0075mm. Considering that the value of surface tension T for an air-water interface is 0.07N/m, calculate from first principles the air entry value of negative pore water pressure for the silty clay (the value of negative pore water pressure at which the silty clay will become unsaturated). Explain what this result means in terms of capillary rise in the silty clay deposit. [15%]

(c) The ground profile at a particular site consists of a layer of sandy silt from part (a), extending from the ground surface to a depth of 2m, overlying a 3m thick layer of the silty clay from part (b). The silty clay has a bulk unit weight γ of 15.0kN/m^3 when it is in a dry condition and 19.1kN/m^3 when it is in a saturated condition. If the water table is at a depth 3m below ground level, calculate the values of vertical total stress σ_v, pore water pressure u and vertical effective stress σ_v' in the sandy silt at the ground surface and at depth 2m. Similarly, calculate the values of vertical total stress σ_v, pore water pressure u and vertical effective stress σ_v' in the silty clay at depths of 2m and 5m below the ground surface. State and justify any assumptions that you make. [25%]

(d) The water table now rises to the ground surface. Calculate new values of σ_v, u and σ_v' in the sandy silt (at the ground surface and at a depth of 2m) and in the silty clay (at depths of 2m and 5m), again justifying any assumptions that you make. [20%]

(e) Will the ground surface experience any settlement/heave as a consequence of the rise of water table mentioned in (d)? Justify you answer from first principles. [25%]

continued

Q2

Figure Q2 shows a scale drawing of the cross-section of a proposed earthdam, which will be used to retain a reservoir with a water depth of 20m. The earthdam will be constructed of a uniform silty clay and it will be founded on very low permeability bedrock. The bulk unit weight of the silty clay is γ of 17.0kN/m^3 when it is in a dry condition and 19.2kN/m^3 when it is in a saturated condition. A gravel toe drain, with a very high permeability, extends a lateral distance of 15m below the dam, as shown in Figure Q2.

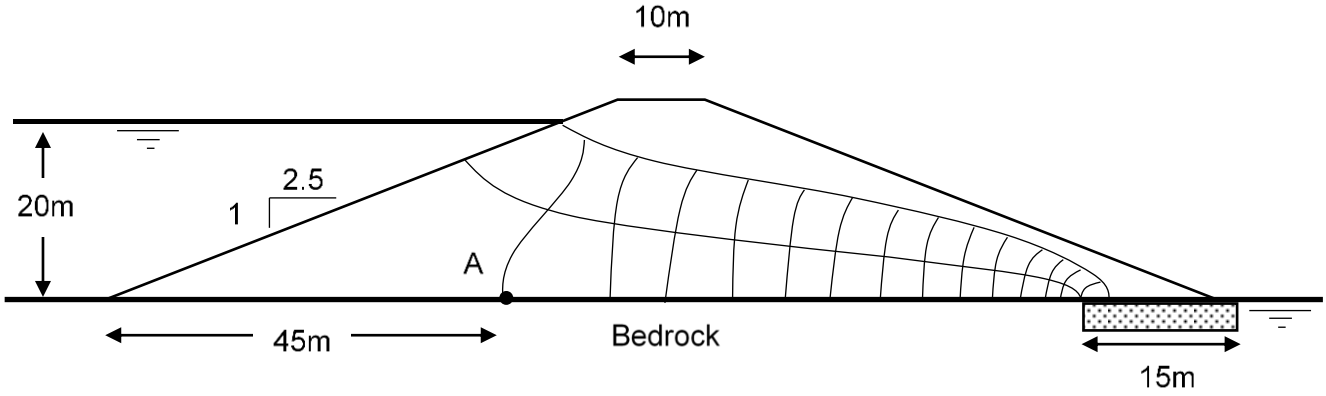


Figure Q2

(a) Based on the flownet sketched in Figure Q2, identify clearly all relevant sections of boundary, and describe the boundary conditions on each of them. [15%]

(b) Use the flownet to calculate the flowrate through the dam (per metre run) if the permeability of the silty clay forming the dam is $5{\cdot}10^{-06}$ m/s. Justify any step in your calculations. [10%]

(c) If the value of permeability given in (b) was unknown, explain and justify one possible method that could be used to estimate the permeability in this particular problem. [10%]

(d) Use the flownet to estimate the value of pore water pressure at point A (see Figure Q2) at the base of the dam. Justify all steps in your calculations. [10%]

(e) To study the compressibility properties of the soil forming the dam, a cylindrical sample of the silty clay is taken from point A (see Figure Q2) and is tested in the lab. If the soil sample in-situ was normally consolidated, calculate the value of its in-situ void ratio. Consider that the compression index of the silty clay is C_c is 0.20, the intercept of its one-dimensional normal compression line e_1 is 1.2 and the swelling index C_s = 0.05. Justify any assumptions that you make. [25%]

(f) From the in-situ stress state at point A (see Figure Q2), the sample of sample is loaded under oedometric conditions to a vertical effective stress of 200 kPa. What is the new value of void ratio? Again, justify any assumptions that you make. [10%]

(g) The sample is finally unloaded to 150 kPa. Calculate the final value of void ratio and the OCR at this final stress state. What is the maximum increment of vertical effective stress that could be applied before the occurrence of plastic strains? [10%]

(h) Sketch in the conventional oedometric plane the complete stress path to which the sample of silty clay has been subjected in the lab. Indicate all the information relevant to compressibility aspects. [10%]

continued

Q3

A retaining wall is constructed on a site to provide stability to a sand fill (see Figure Q3). The site geotechnical engineer wants to ensure that the retaining wall is stable against overturning according to Eurocode 7 (Case C). The dimensions of the wall are: H=4.5m, B=3.1m and t=0.5m as indicated in Figure Q3. The fill material consists of sand with a unit weight γ=19kN/m^3 and a friction angle φ'=25°. The unit weight of the reinforced concrete in the wall is 25kN/m^3. The ground water level is at foundation level, as shown in Figure Q3. The wall carries a variable uniform surcharge q of 15 kPa.

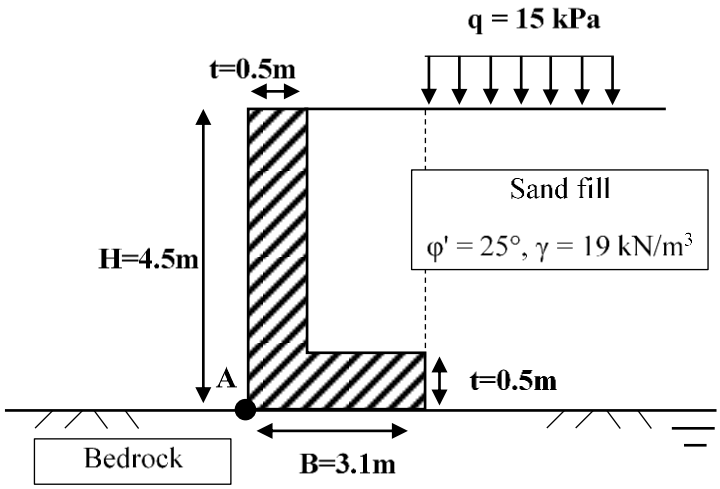


Figure Q3

(a) Assuming no friction between the sand fill and the wall, calculate the active earth pressure coefficient, K_a. [10%]
(b) Sketch and calculate the horizontal components of pressures and calculate the corresponding thrusts acting on the wall. [10%]
(c) Sketch and calculate the vertical components of pressures and calculate the corresponding thrusts acting on the wall. [10%]
(d) Calculate the stability against overturning according to Eurocode 7 (Case C) by ensuring that moments about point A are stable. [20%]
(e) Assuming a friction between the sand fill and the wall of φ_w'=10°, calculate the active earth pressure coefficient, K_a. [10%]
(f) Calculate the stability against overturning in this new situation and according to Eurocode 7 (Case C) by ensuring that moments about point A are stable. [20%]
(g) Compare and discuss the results obtained in (d) and (f). [10%]
(h) After an intense rainfall, the water level rises to the top of the ground surface. Explain how this raise in water level will influence the stability of the retaining wall. Justify your answer. [10%]

continued

Q4

A circular footing is founded on a dry stiff till with a characteristic undrained shear strength C_{uk}=73kPa, characteristic angle of friction φ'_k=30° and characteristic cohesion c_k'=0kPa. The water table is at bedrock (see Figure Q4). The footing supports a vertical permanent load V_{1k}=850kN (not including self-weight) and a vertical variable load of V_{2k}=500kN, both applied centrally to the footing (so that eccentricity can be ignored). The unit weight of the dry stiff till is γ_k=18.5kN/m^3 and the unit weight of the reinforced concrete is γ_{kc}=25kN/m^3.

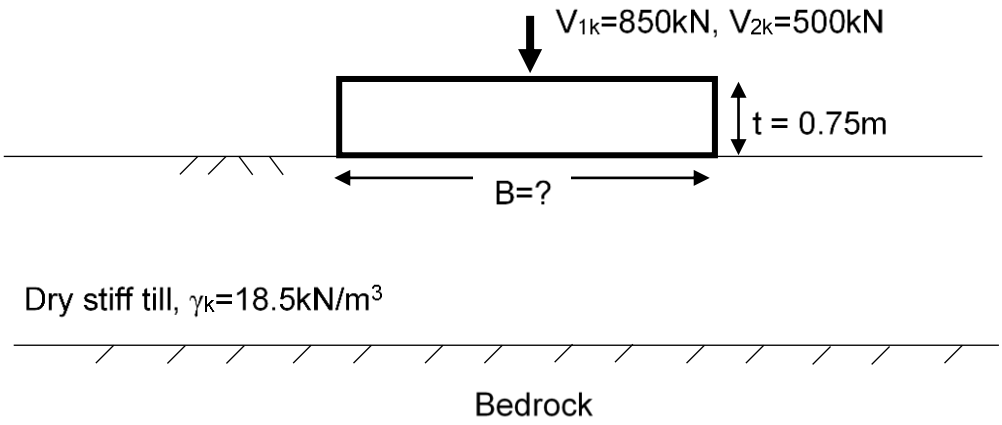


Figure Q4

Following the next steps, find the minimum value of the foundation diameter B that ensures the safety of the footing in accordance with the Eurocode 7 (Case C) for short and long term conditions.

(a) Express the design bearing pressure q_{Ed} in terms of B. [20%]
(b) Calculate the design bearing resistance for short term conditions, $q_{Rd,U}$. [20%]
(c) What is the minimum value of the foundation diameter B that ensures the safety of the footing under short term conditions? [20%]
(d) Express the design bearing resistance for long term conditions, $q_{Rd,D}$, in terms of B. [20%]
(e) Assuming the value of B found in c), is the footing stable in the long term? Is this an expected outcome? Justify your answer. [20%]

continued

Phase relationships

$e=\frac{V_v}{V_s}$	$S_r=\frac{V_w}{V_v}$	$v=1+e$	$n=\frac{e}{1+e}$	$w=\frac{M_w}{M_s}$
$\rho_s=G_s\rho_w$	$wG_s=S_re$	$\rho=\frac{M_s+M_w}{V_s+V_v}$	$\rho=\frac{(G_s+S_re)\rho_w}{1+e}$	$\gamma=\rho g$
$\gamma=\frac{(G_s+S_re)\gamma_w}{1+e}$	$\gamma'=\gamma_{sat}-\gamma_w$	$\rho_d=\frac{G_s\rho_w}{1+e}$	$\rho_d=\frac{\rho}{1+w}$	$\gamma_w\approx 10$ kN/m^3

Stresses and pore pressures in soils:

$\sigma_v=\rho gz=\gamma z$	$u_w=\rho_w gz_w=\gamma_w z_w$	$\sigma_v'=\sigma_v-u$	$u_a-u_w=\frac{2T}{R}$	$h_c=\frac{2T}{\gamma_w R}$

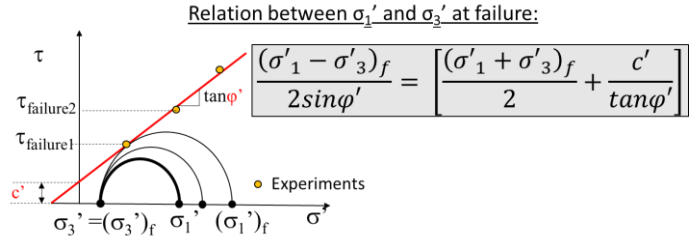
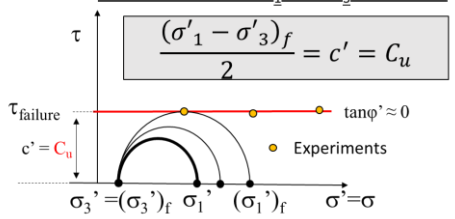
Elastic stress-strain relations:

The volumetric strain is: $\Delta\varepsilon_v=\Delta\varepsilon_1+\Delta\varepsilon_2+\Delta\varepsilon_3=\Delta\varepsilon_x+\Delta\varepsilon_y+\Delta\varepsilon_z$

$\Delta\varepsilon_x=\frac{1}{E'}\left(\Delta\sigma_x'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{xy}=\frac{\Delta\tau_{xy}}{G'}$	--
$\Delta\varepsilon_y=\frac{1}{E'}\left(\Delta\sigma_y'-\nu'\Delta\sigma_x'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{yz}=\frac{\Delta\tau_{yz}}{G'}$	$K'=\frac{E'}{3(1-2\nu')}$
$\Delta\varepsilon_z=\frac{1}{E'}\left(\Delta\sigma_z'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_x'\right)$	$\Delta\gamma_{zx}=\frac{\Delta\tau_{zx}}{G'}$	$G'=\frac{E'}{2(1+\nu')}$
$\Delta\varepsilon_v=\frac{\Delta p'}{K'}$		$\Delta\varepsilon_q^e=\frac{\Delta q}{3G'}$
$p'=\frac{\sigma'_1+\sigma'_2+\sigma'_3}{3}$		$q=\left[\frac{(\sigma_1'-\sigma_2')^2+(\sigma_2'-\sigma_3')^2+(\sigma_3'-\sigma_1')^2}{2}\right]^{1/2}$

where σ'_1, σ'_2 and σ'_3 are the principal effective stresses (i.e. $\sigma'_1\geq\sigma'_2\geq\sigma'_3$)

Failure envelopes:

$\tau=c'+\sigma'\tan\varphi'$ (Mohr-Coulomb)	$\tau=C_u$ (Tresca)
Relation between σ_1' and σ_3' at failure: $\frac{(\sigma'_1-\sigma'_3)_f}{2\sin\varphi'}=\left[\frac{(\sigma'_1+\sigma'_3)_f}{2}+\frac{c'}{\tan\varphi'}\right]$ 	Relation between σ_1' and σ_3' at failure: $\frac{(\sigma'_1-\sigma'_3)_f}{2}=c'=C_u$ 

Ground water flow:

$h=z+\frac{u}{\rho g}=z+\frac{u}{\gamma_w}$	$q_x=Aki_x=Ak\left(-\frac{\partial h}{\partial x}\right)$	$v_x=ki_x$	$v_{true,x}=v_x\frac{1}{n}$

Constant head permeameter:	Falling head permeameter:	Pumping test:
$q=Ak\frac{h_1-h_2}{L}$	$\ln\left(\frac{h_0}{h}\right)=\frac{Ak}{aL}t$	$\left(H^2-h^2\right)=\frac{q}{\pi k}\left(\ln R_0-\ln r\right)$

Laplace's Equation (3D)	Laplace's Equation (2D, flownets)	
$\frac{\partial^2 h}{\partial x^2}+\frac{\partial^2 h}{\partial y^2}+\frac{\partial^2 h}{\partial z^2}=0$	$\delta h=\frac{\Delta h}{N_d}$	$q=k\frac{N_f}{N_d}\Delta h;\ \ u=\gamma_w\left(h-z\right)$

continued

Critical hydraulic gradient:	$i_c = \dfrac{\gamma - \gamma_w}{\gamma_w}$

1D flow through an unconfined aquifer:	1D flow through a confined aquifer:
$q_x = \dfrac{k\left(H_1^2 - H_2^2\right)}{2L}$	$q_x = Aki_x = -Ak\dfrac{\partial h}{\partial x} = -k\dfrac{H_2 - H_1}{L}D$
$h = \sqrt{\dfrac{\left(H_2^2 - H_1^2\right)}{L}x + H_1^2}$	$h = \dfrac{H_2 - H_1}{L}x + H_1$

1D straining (oedometric test conditions):

1D normal compression line:	1D swelling line:
$e = e_1 - C_c \log_{10} \sigma'_v$	$e = e_2 - C_s \log_{10} \sigma'_v$

$OCR_{1D} = \dfrac{max.\ past\ value\ of\ \sigma'_v}{current\ value\ of\ \sigma'_v} = \dfrac{\sigma'_{vc}}{\sigma'_v}$	$\sigma'_h = K_0 \sigma'_v$	$K_{0nc} \approx 1 - \sin\varphi'$	$K_0 \approx (1 - \sin\varphi')\sqrt{OCR}$

Settlement of a very wide foundation:		
$\dfrac{\Delta H}{H_0} = \dfrac{e_0 - e_1}{1 + e_0}$	$\Delta H = \dfrac{H_0}{1 + e_0}\left[C_s \log_{10}\dfrac{\sigma'_{vc}}{\sigma'_{v0}} + C_c \log_{10}\dfrac{\sigma'_{v1}}{\sigma'_{vc}}\right]$	$\Delta\varepsilon_v = m_v \Delta\sigma'_v$

1D consolidation:		
$\dfrac{\partial u_e}{\partial t} = c_v\left(\dfrac{\partial^2 u_e}{\partial z^2}\right)$	$c_v = \dfrac{k}{m_v \gamma_w}$	$T_v = \dfrac{c_v t}{d^2}$

Axisymmetric conditions (triaxial test conditions):

$\Delta\varepsilon_v = \Delta\varepsilon_a + 2\Delta\varepsilon_r$	$\Delta\varepsilon_q = \dfrac{2}{3}\left(\Delta\varepsilon_a - \Delta\varepsilon_r\right)$
$p' = \dfrac{1}{3}\left(\sigma_a' + 2\sigma_r'\right)$	$q = \sigma_a - \sigma_r$

where σ'_a and σ'_r are the principal effective axial and radial stresses respectively

Iso-normal compression line:	Iso-swelling line:
$v = N - \lambda \ln p'$	$v = v_\kappa - \kappa \ln p'$

Critical states:	
$q = Mp'$	$v = \Gamma - \lambda \ln p'$

where $\Gamma = N - (\lambda - \kappa)\ln(2)$

Relation between friction angle and critical state gradient:	$M = \dfrac{q}{p'} = \dfrac{6\sin\varphi'}{3 - \sin\varphi'}$

Overconsolidation ratio (OCR in terms of p')	$OCR = \dfrac{max.\ past\ value\ of\ p'}{current\ value\ of\ p'} = \dfrac{p'_0}{p'}$

Partial factors Eurocode 7 & factor of safety: (Bond & Harris, 2008; *Decoding EC7*)

$q_{Ed} \le q_{Rd}$ where, q_{Ed} is the design pressure and q_{Rd} is the design resistance	CASE	ACTIONS			Soil properties	
		Permanent		Variable	tan φ', c'	C_u
		Unfavourable	Favourable	Unfavourable		
	C	1.00	1.00	1.30	1.25	1.4

continued

Factor of safety, FoS:	$FoS = \frac{\text{available shear force, } T_f}{\text{allowable working load, } T}$; $FoS = \frac{\text{available bearing capcity, } \tau_f}{\text{allowable bearing capacity, } \tau}$

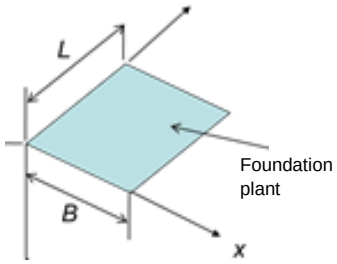
Shallow foundation settlement, δ:

Cohesionless soils:

Burland & Burbage:

$$\delta\ (mm) = f_s \cdot f_l \cdot f_t \cdot \left[\left(q'_n - \frac{2}{3}p'_0\right) \cdot B^{0.7} \cdot I_c\right]$$

where,



$f_s = \left(\frac{1.25 \cdot L/B}{\frac{L}{B} + 0.25}\right)^2$	$f_l = \frac{H}{z_l}\left(2 - \frac{H}{z_l}\right)$	$f_t = 1 + R_3 + R \cdot log\left(\frac{t}{3}\right)$

Cohesive soils: $\delta = \delta_i + \delta_c$ where,

$\delta_i\ (mm) = \mu_1 \cdot \mu_0 \cdot q_n \frac{B}{E_u}$	$\delta_c\ (mm) = \mu_g \cdot \delta_{oed}$ where, $\delta_{oed}\ (mm) = m_v \cdot \sigma'_v \cdot H$

Bearing capacity:

Brinch-Hansen (drained all soils)

$$q_f = c' \cdot N_c \cdot s_c + p'_{ov} \cdot N_q \cdot s_q + \frac{1}{2} \cdot \gamma' \cdot B \cdot N_y \cdot s_y$$

where,

$N_q = e^{\pi \tan \varphi'} \cdot \left(\tan(45 + \varphi'/2)\right)^2$	$s_q = 1 + (B'/L') \cdot \sin \varphi'$ (rectangular); $s_q = 1 + \sin \varphi'$ (circular and square)
$N_c = (N_q - 1) \cdot \cot \varphi'$	$s_c = (s_q \cdot N_q - 1)/(N_q - 1)$ (all)
$N_y = 2 \cdot (N_q - 1) \cdot \tan \varphi'$	$s_y = 1 - 0.3 \cdot (B'/L')$ (rectangular; $s_y = 0.7$ (circular and square)

depth factors are not permitted within EC7; load inclination, base and ground factors ignored.

Fine-grained soils (undrained):

$q_f = 5.14 \cdot C_u \cdot s'_c$ where,

$s'_c = 1 + 0.2 \cdot \left(\frac{B'}{L'}\right)$ (rectangular shape; for circular and square shape B'=L')

load inclination, base and ground factors are ignored.

Bearing capacity in piles:

$$Q_u = Q_s + Q_b - W_p$$

Cohesionless soils:	Cohesive soils:
$Q_s = q_s \cdot A_s = A_s \cdot k_s \cdot \sigma_{v0} \cdot \tan \delta$	$Q_s = q_s \cdot A_s = \alpha \cdot C_u \cdot A_s$
$Q_b = q_b \cdot A_b = N_q \cdot \sigma'_{v0} \cdot A_b$	$Q_b = q_b \cdot A_b = N_c \cdot C_{ub} \cdot A_b$

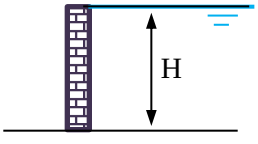
Passive and active pressures:

$\sigma'_h = K_a \cdot \sigma'_v$ (active)	$\sigma'_h = K_b \cdot \sigma'_v$ (passive)

RANKINE'S theory: H high of the layer of soil behind the wall retaining wall, φ' is the friction angle and c' is the cohesion

In the following, $P_a = P_a' + U$ and $P_p = P_p' + U$,

where U is the resultant of the pore water pressure.



$K_a = \tan^2\left(\frac{\pi}{4} - \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1+\sin\varphi'}\right)^2 = \frac{1-\sin\varphi'}{1+\sin\varphi'}$	$K_p = \tan^2\left(\frac{\pi}{4} + \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1-\sin\varphi'}\right)^2 = \frac{1+\sin\varphi'}{1-\sin\varphi'}$

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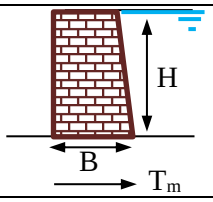
Cohesionless soils:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u$	$\sigma_h' = \sigma_v' K_a - 2c'\sqrt{K_a}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{1}{2}(\gamma H - 2C_u)\left(H - \frac{2C_u}{\gamma}\right)$	$P_a' = \frac{K_a \gamma'}{2}\left(H - \frac{2c'}{\gamma'\sqrt{K_a}}\right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = 2C_u + \sigma_v$	$\sigma_h' = \sigma_v' K_p + 2c'\sqrt{K_p}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}(\gamma H + 4C_u)$	$P_p' = \frac{1}{2} K_p \gamma' H^2 + 2c'\sqrt{K_p} H$

COULOMB'S theory: Friction and adhesion between soil and wall are φ_w and c_w, resp.

$$K_a = \left(\frac{\cos\varphi'}{1+\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}}\right)^2 \frac{1}{\cos\varphi_w'} \qquad K_p = \left(\frac{\cos\varphi'}{1-\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}}\right)^2 \frac{1}{\cos\varphi_w'}$$

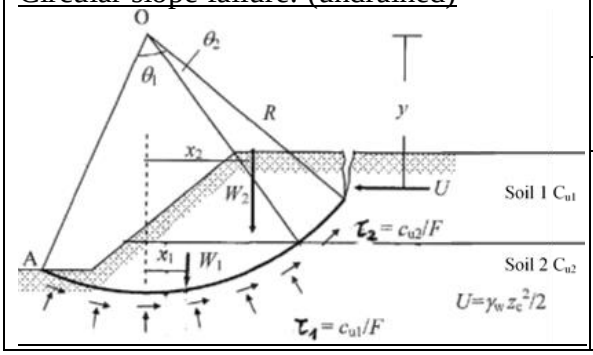
Cohesionless:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u\sqrt{1+\frac{C_w}{C_u}}$	$\sigma_h' = K_{a1}\sigma_v' - c'K_{a2}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{\gamma}{2}\left(H - \frac{2C_u}{\gamma}\sqrt{1+\frac{C_w}{C_u}}\right)^2$	$P_a' = \frac{K_{a1}\gamma'}{2}\left(H - \frac{c'K_{a2}}{\gamma' K_{a1}}\right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = \gamma z + 2C_u\sqrt{1+\frac{C_w}{C_u}}$	$\sigma_h' = \gamma z K_{p1} + c'K_{p2}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}\left(\gamma H + 4C_u\sqrt{1+\frac{C_w}{C_u}}\right)$	$P_p' = \frac{1}{2} K_{p1}\gamma' H^2 + c'K_{p2}H$

where,

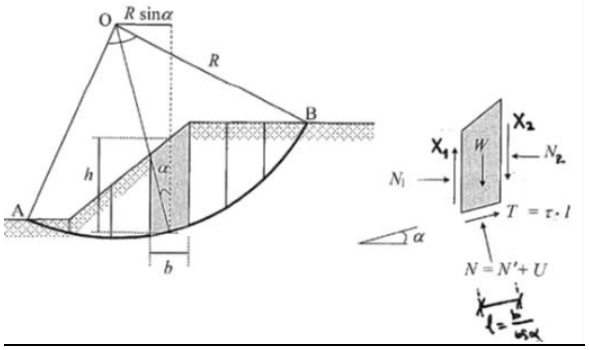
$K_{a1} = K_a$	$K_{p1} = K_p$
$K_{a2} = \frac{\cos\varphi'\left(2\sqrt{1+c_w'/c'} + (2+c_w'/c')\sin\varphi'\right)}{\left(1+\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}\right)^2\cos\varphi_w'}$	$K_{p2} = \frac{\cos\varphi'\left(2\sqrt{1+c_w'/c'} - (2+c_w'/c')\sin\varphi'\right)}{\left(1-\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}\right)^2\cos\varphi_w'}$
Cantilever wall in sand:	$FoS = \frac{N'\tan\varphi_w'}{T_m} = \frac{(W-U)\tan\varphi_w'}{T_m}$
Gravity wall in sand:	$FoS = \frac{N\tan\varphi_w'}{T_m}$
Gravity wall in clay: 	(Undrained) $FoS = \frac{C_w B}{T_m}$ / (Drained) $FoS = \frac{N'\tan\varphi_w' + c_w' B}{T_m}$

continued

Slope stability: (Note: for undrained strength in infinite slope take c'=C_u and φ'=0)

Infinite slope with parallel flow: H: layer thickness, m: position of water table from bedrock, β: slope angle; γ_1 (dry soil; m=0), γ_2 (saturated soil; m=1).	$FoS = \dfrac{c' + H\left(\gamma_1(1-m) + m(\gamma_2 - \gamma_w)\right)\cos^2\beta\tan\varphi'}{H\left((1-m)\gamma_1 + m\gamma_2\right)\cos\beta\sin\beta}$
Vertical cut (dry soil):	$FoS = \dfrac{4C_u}{H\gamma\sin 2\alpha}$ (undrained)
	$FoS = \dfrac{4(c' + \sigma'\tan\varphi')}{H\gamma\sin 2\alpha}$ (drained)
Circular slope failure: (undrained) 	$FoS = \dfrac{R^2\theta C_u}{Wx}$
	$FoS = \dfrac{R^2\theta C_u}{Wx + Uy}$ (with cracks)
	$FoS = \dfrac{(\theta_1 C_{u1} + \theta_2 C_{u2})R^2}{W_1 x_1 + W_2 x_2 + Uy}$ (layered soil)

$\sin 2\alpha = 2\sin\alpha\cos\alpha$, $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$ and arc-length: $AB = R\theta$ (θ is in radians).

Circular: (drained): 	$FoS = \dfrac{\sum_{i=1}^{n} l\left(c' + \sigma'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} = \dfrac{\sum_{i=1}^{n}\left(lc' + N'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$
Modified Bishop's b: width of the slice, l: length of the slice, α: inclination of the slice to the horizontal direction.	$FoS = \dfrac{\sum_{i=1}^{n}\left(bc' + \left(W - U\cos\alpha\right)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} M(\alpha)$ where, $M(\alpha) = \dfrac{1}{\cos\alpha\left[1 + \dfrac{\tan\alpha\tan\varphi'}{FoS}\right]}$
Fellenius:	$FoS = \dfrac{\sum_{i=1}^{n}\left(lc' + \left(W\cos\alpha - ul\right)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$

END